OVERLAP

A voice agent that knows "haan" doesn't mean "stop".

Arsh · built on Gnani Prisma, Evon and Timbre · repo README has the run steps

The problem

I called a support line about a shirt I'd bought and it turned out to be an AI voice agent. It was fine until I said "okay" while it was talking. It stopped mid-word, went quiet, and then started the whole answer over.

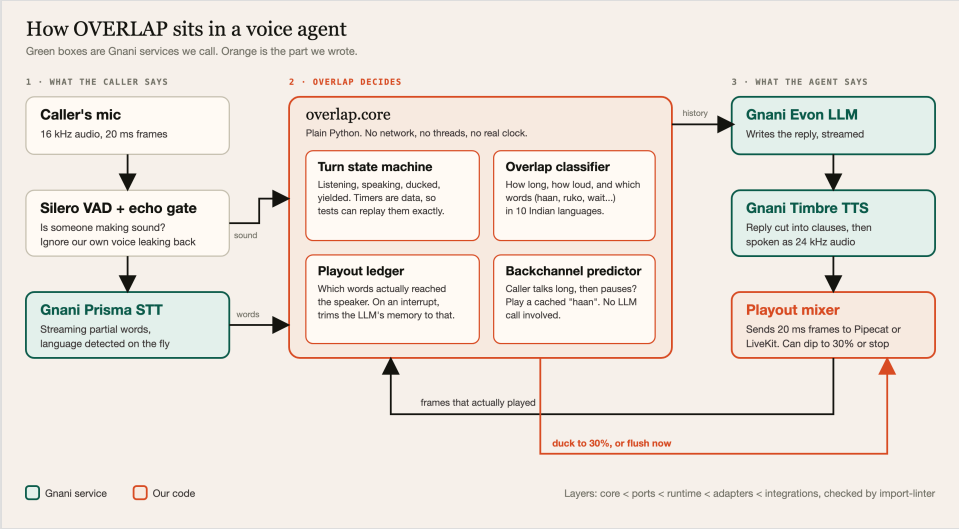
That's how most voice agents work. The voice activity detector hears you, so the agent's audio gets killed. A "hmm", a cough, a "haan" all count as you taking over, and then both people say "no no, go ahead". There's a quieter bug too. The LLM writes much faster than the voice speaks, so when you do interrupt, it still believes it said everything and next turn acts like you heard things you never did.

What we built

OVERLAP sits between the caller's mic and the agent's speaker in a Gnani pipeline and does three things.

1. **Turn down, don't stop.** When the caller talks over the agent, the volume drops to 30% inside one 20 ms frame and the agent keeps going. About a quarter of a second later we know what it was. "Haan" or "okay": volume comes back and the sentence finishes. "Ruko" or "wait": the agent stops.
2. **Remember only what was said.** We track which words actually reached the speaker. On an interruption the LLM's history is cut to that point.
3. **Nod along.** When the caller explains something long and pauses, the agent drops in a short "haan" or "achha" from a small cache of pre-made Timbre clips. No LLM call, so it's on time.

It has word lists for 10 languages (Hindi, Tamil, Telugu, Kannada, Malayalam, Marathi, Bengali, Gujarati, Punjabi, Indian English) and plugs into a Pipecat pipeline in about ten lines. There's also a LiveKit transport.



Does it work?

We replayed 240 labelled cases across the 10 languages through the real classifier and state machine.

Overall accuracy	Held-out (60 cases, never tuned on)	Backchannels that stopped the agent	Real interruptions missed	Decision time (p95)
235 / 240 (97.9%)	58 / 60 (96.7%)	0	0	321 ms

On the held-out set the only two errors were side-talk (someone talking to someone else in the room) read as an interruption. Being straight about the gaps: this is replayed test data, not a live call with strangers, and the blind listening test with real people is built but not run yet.